



Gymnosperms & Angiosperms

Seed Plants and Flowering Plants

What types of plants are there?



All Seed Plants are Vascular Plants



Spore vs. Seed



Spores:

- nonvascular plants-
 - water needed for fertilization
 - reproduces with spores
- vascular seedless plants
 - water needed for fertilization
 - reproduces with spores

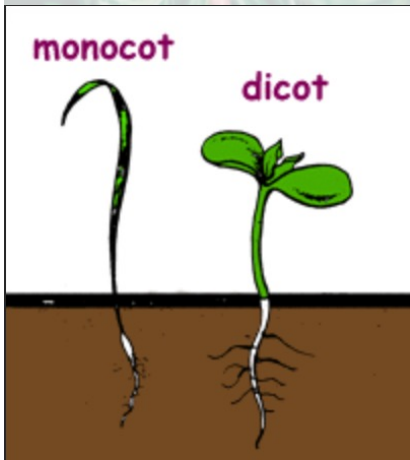
Seeds:

- vascular seeded plants
 - gymnosperms- cones
 - angiosperms
 - make pollen and ovules (ovaries)
 - once fertilized, the ovules produce fruit
 - fruit houses/protects seeds

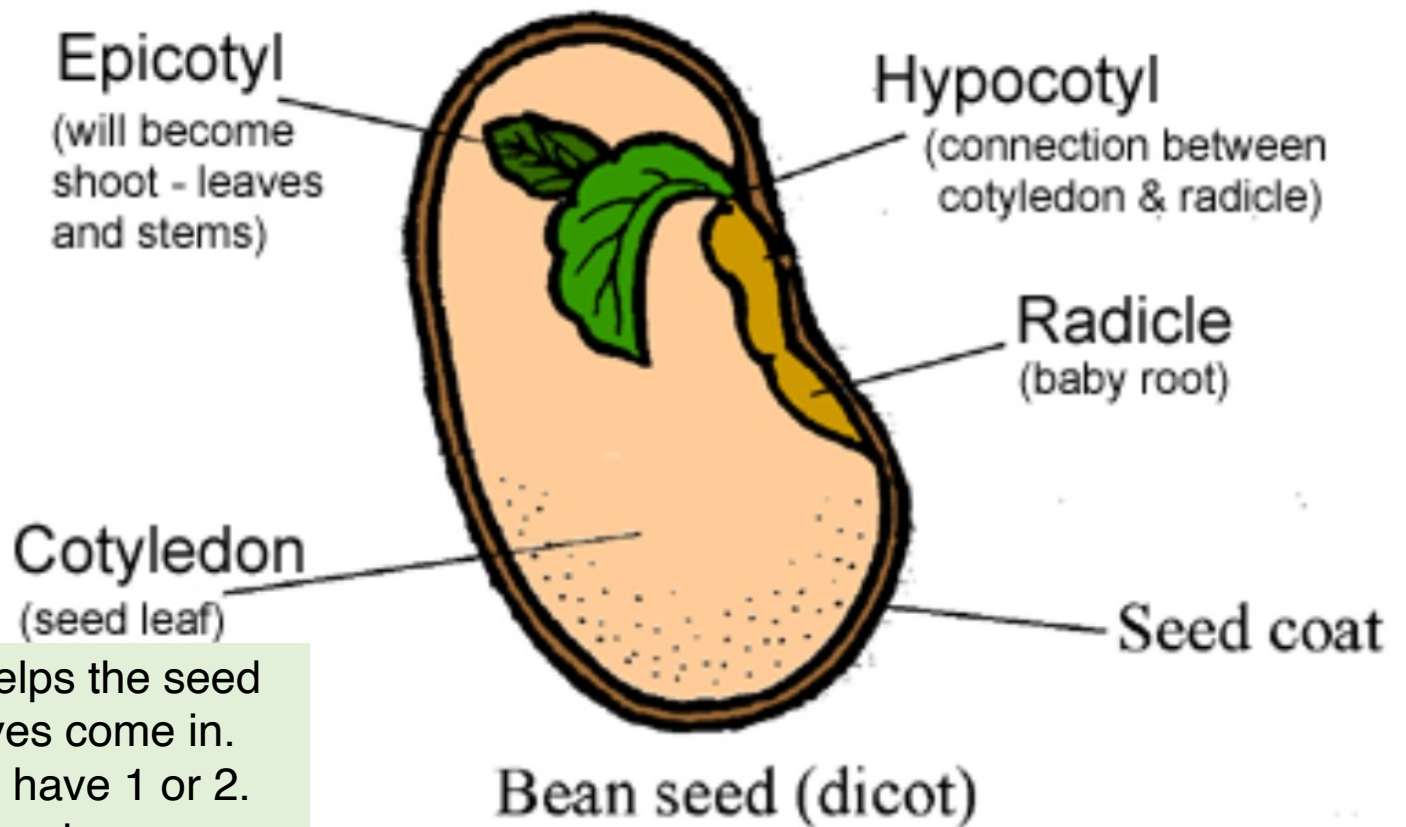
What is a seed?

Seeds help a plant get started when it lands in the right conditions

Look at the parts of this bean seed.
How will the seed help a new bean plant to grow?



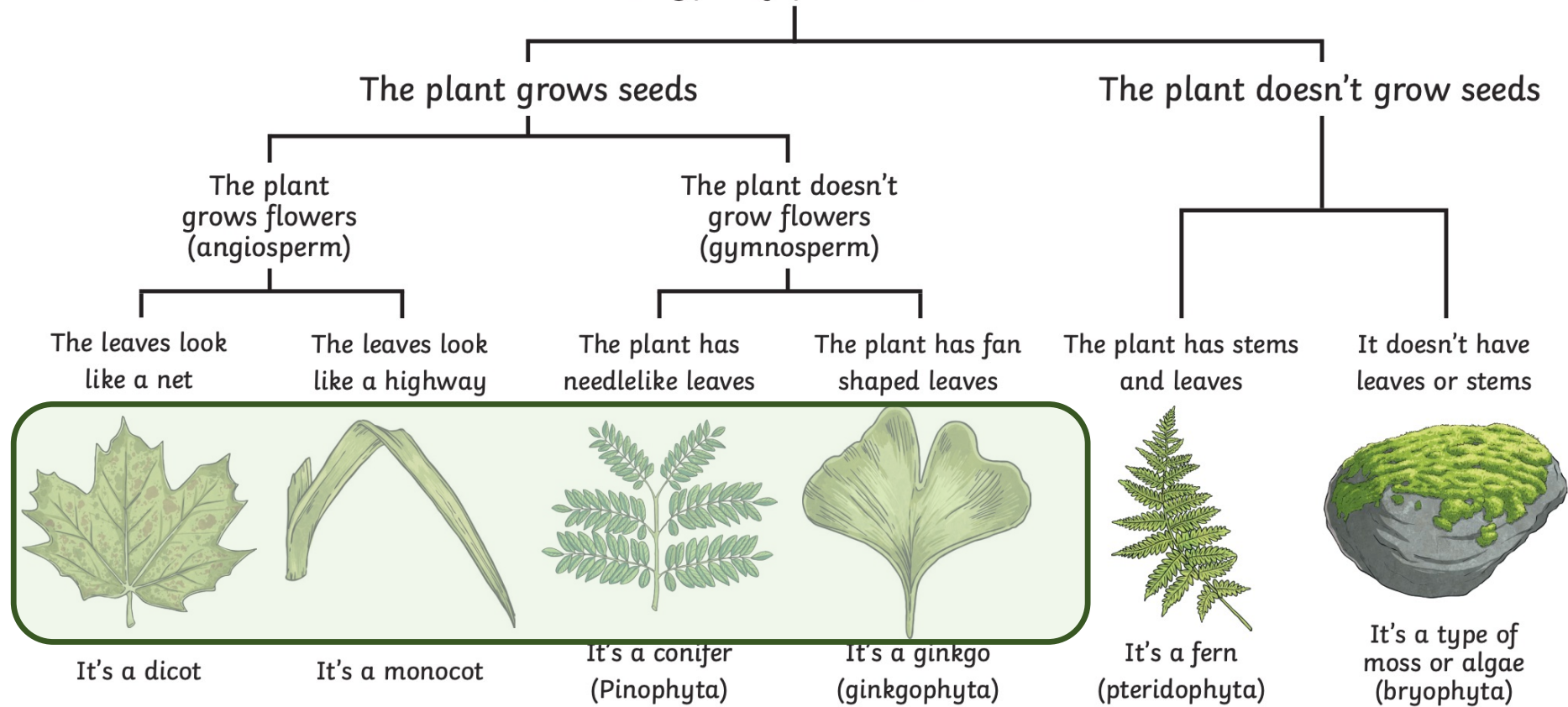
The cotyledon helps the seed grow before leaves come in. Flowering plants have 1 or 2. Gymnosperms can have more.



Which Kinds of Plants Make Seeds?

Dichotomous Key for Plants

What type of plant is that?



Gymnosperms

- Make seeds but not flowers
- The seeds are exposed, with no fruit covering them
- Can use wind to spread seeds
- First plants to make forests on earth



Examples of Gymnosperms

conifers

(cone bearing)



cycads

(tropical and palmlike)



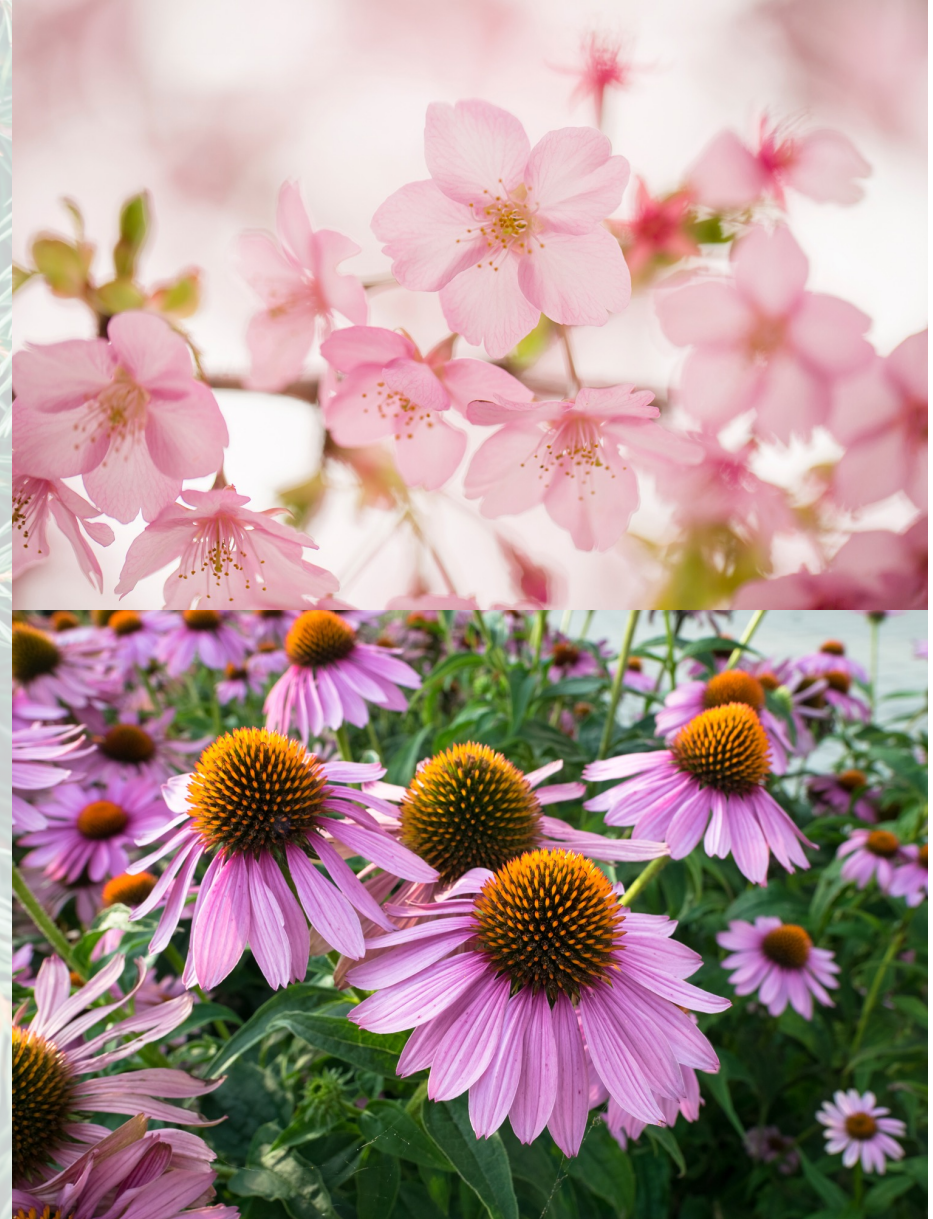
ginkgo

(maidenhair tree)

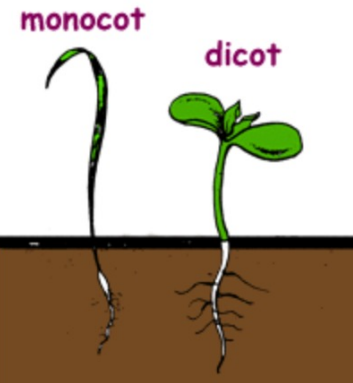


Angiosperms

- Most recent and diverse group of plants
- Includes all flowering plants
- Most use animals to pollinate
- Seeds form in the fertilized flower, which becomes the fruit
- Seeds are surrounded by a type of fruit, but not always edible
- Highly adaptive and can use many strategies to survive



2 Main Types: Monocots & Dicots



Monocots



One cotyledon



Veins usually parallel



Vascular bundles usually complexly arranged



Fibrous root system



Floral parts usually in multiples of three

Embryos

Leaf venation

Stems

Roots

Flowers

Dicots



Two cotyledons



Veins usually netlike



Vascular bundles usually arranged in ring

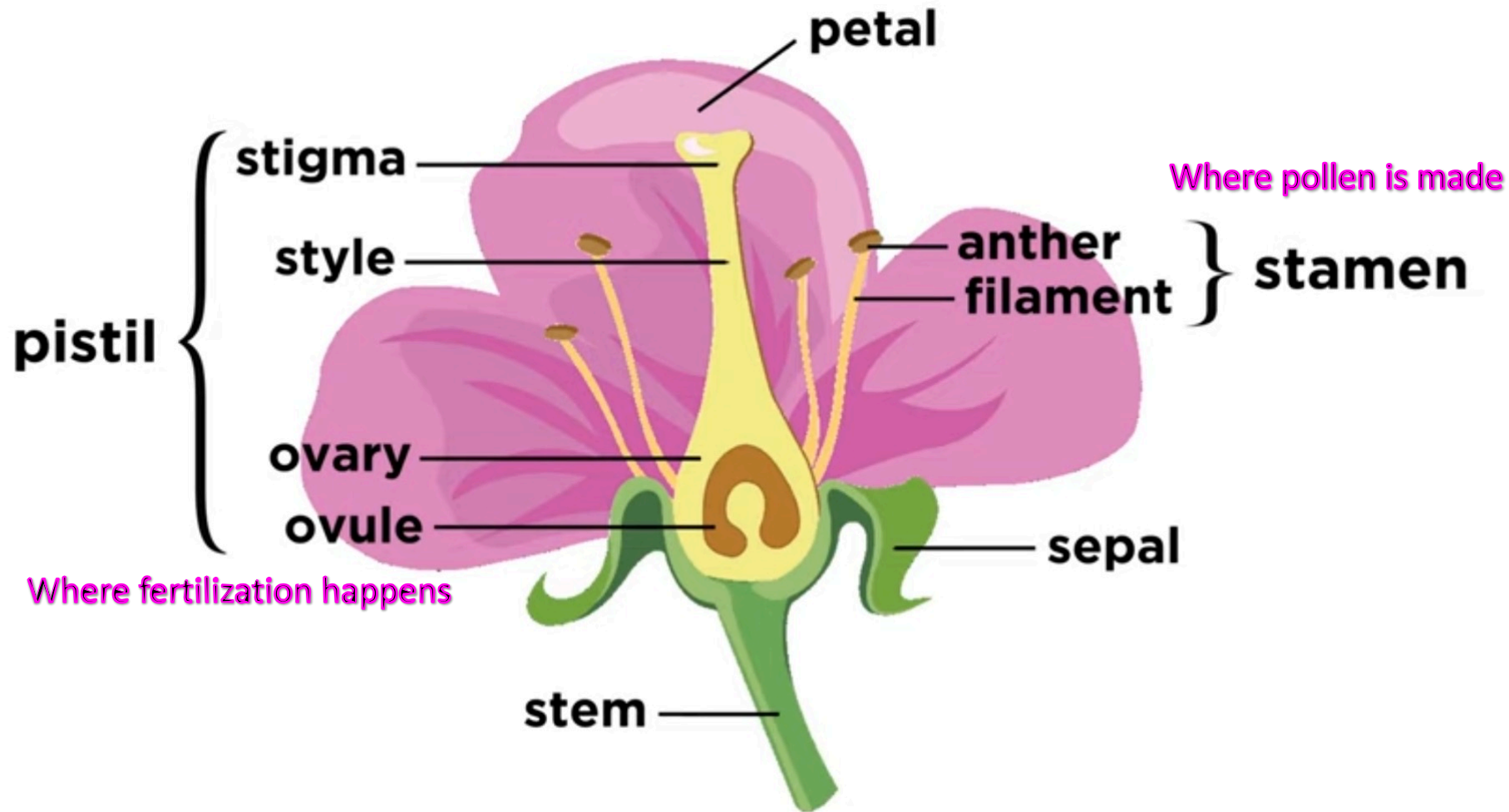


Taproot usually present

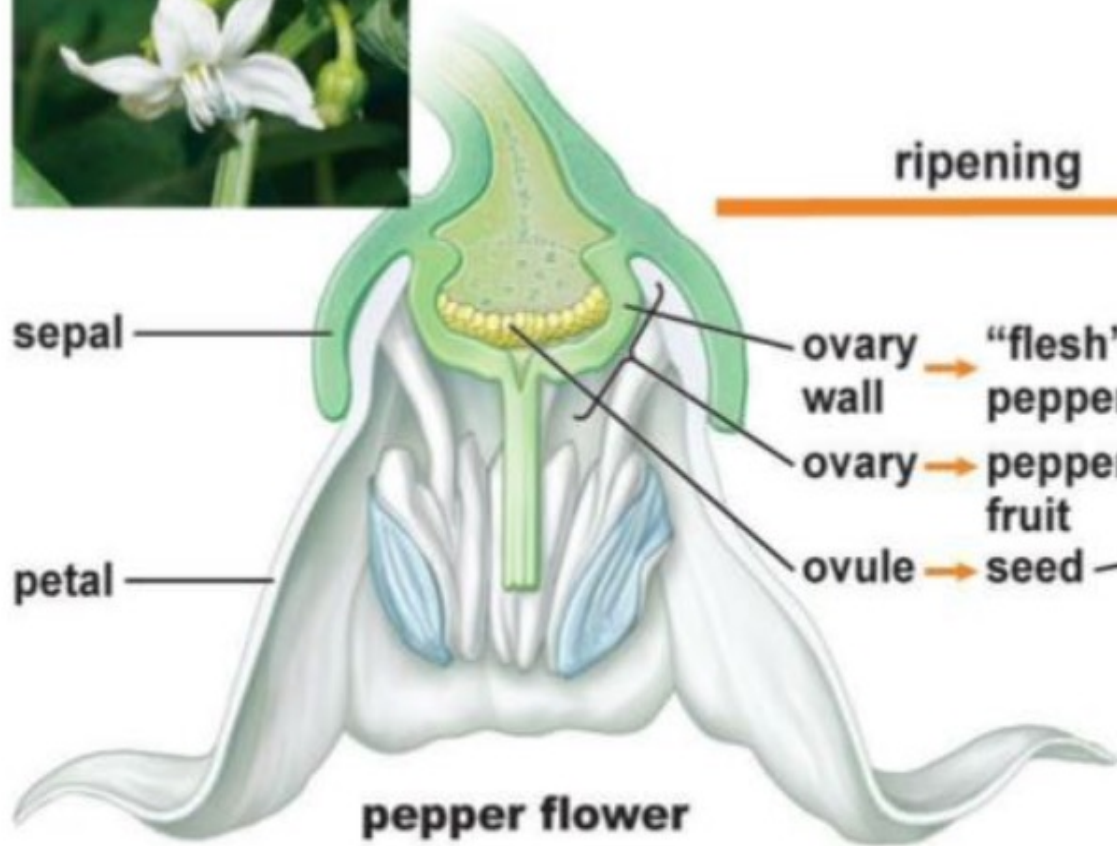


Floral parts usually in multiples of four or five

Parts of a Flower



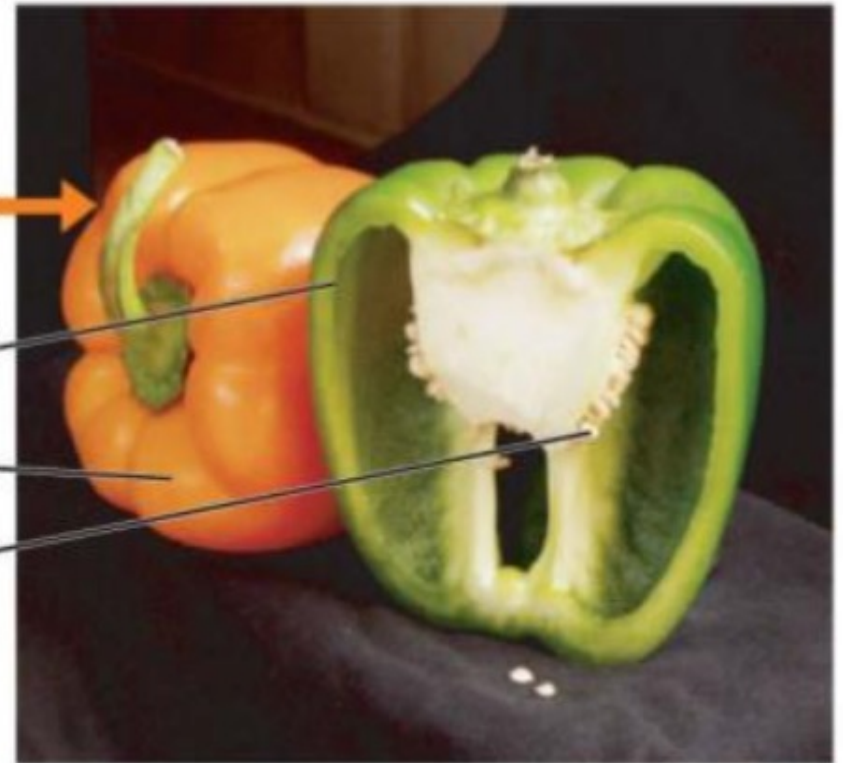
From Flowers to Fruit



"flesh" of
pepper

pepper
fruit

seed



pepper fruits

How Seeds Travel

by the wind



milkweed



dandelion



maple

by animals



beggar-ticks



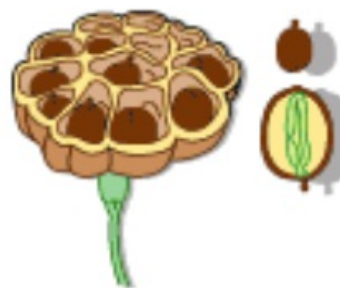
sandbur



blackberry

by water

lotus



cattail



coconut

by bursting

violet



jewelweed



witch hazel

by humans

bean



wheat



cherry



